Computer Science - Graduate Level Assessment 11 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. A starvation-free job-scheduling policy guarantees that no job waits indefinitely for service. Which of the following job-scheduling policies is starvation-free?
   (a) Round-robin
   (b) Priority queuing
   (c) Shortest job first
   (d) Youngest job first
2. Which of the following decimal numbers has an exact representation in binary notation?
   (a) 0.1
   (b) 0.2
   (c) 0.3
   (d) 0.5
3. Which of the following is NOT a property of bitmap graphics?
   (a) Fast hardware exists to move blocks of pixels efficiently.
   (b) Realistic lighting and shading can be done.
   (c) All line segments can be displayed as straight.
   (d) Polygons can be filled with solid colors and textures.
4. Which of the following is usually NOT represented in a subroutine's activation record frame for a stack-based programming language?
   (a) Values of local variables
   (b) A heap area
   (c) The return address
   (d) Stack pointer for the calling activation record
5. Which of the following are most vulnerable to injection attacks?
   (a) Session IDs
   (b) Registry keys
   (c) Network communications
   (d) SQL queries based on user input

True / False (10 marks)

Answer True or False

6. True or False: Mutation-based fuzzing works by making small mutations to the target program to induce faults.
7. True or False: The AH Protocol provides source authentication and data integrity, but it does not ensure privacy for the transmitted data.
8. True or False: Message integrity ensures that data arrives at the receiver exactly as it was sent, without alterations during transmission.
9. True or False: When the STP solver times out on a constraint query, EXE assumes the query is not satisfiable and stops executing the path.
10. True or False: Using busy-waiting for an asynchronous event is unreasonable in a time-sharing system due to potential CPU resource wastage.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to find the slope of a line.
12. Write a function to convert snake case string to camel case string.

End of Paper